# Functional Representation Lemma and Its Applications

Student: Serhat Emre Coban, Advisor: Yanina Shkel, Co-advisor: Emre Telatar
Mathematics of Information Lab (MIL)

***Abstract*—This project is about Strong Functional Representation Lemma (SFRL)[1], and its applications to one-shot achievability problems in Information Theory. For given random variables $X$ and $Y$, functional representation means finding a random variable $Z$, and a deterministic function $g$ such that $Z$ is independent of $X$ and $Y$ can be represented as $g(X, Z)$. We can think of SFRL as a tool that provides functional representation with an Information Theoretic bound on the conditional entropy of $Y$ given $Z$ (i.e. $H(Y|Z) < I(X,Y) + log(I(X,Y)+1)+4$). In my project,first, I went through the proof and applications of SFRL to several coding problems[1] from channel simulation to lossy source coding. Then, I started to Poisson Matching Lemma [2] which is extended version of the work done in [1]. In this report, I summarized these works and presented my understandings about them.**

***Index Terms*—Strong Functional Representation Lemma, Poisson Matching Lemma, Achievability, Channel Simulation, Lossy Source Coding**

## I. INTRODUCTION

STRONG functional representation lemma, which is introduced in [1] with its applications to some fundamental coding problems in Information Theory, is a technique that provides a functional representation for a given random variable $Y$ in terms of a given random variable $X$, and a random variable of our choice $Z$ which is independent of $X$. There are several other methods that provides functional representation other than SFRL, but what makes it unique is that it actually helps us to make this functional representation in a way that $Z$ is most informative about $Y$ (i.e. minimizes the conditional entropy $H(Y|Z)$). In section II, I am going present some functional representation methods with examples and natural limits for functional representation. In section III, I will present the Strong Functional Representation Lemma (and give some idea about its proof) which is the method that we use for making $Z$ as informative as possible about $Y$ in the functional representation problem. The importance of making $Z$ as informative as possible about $Y$ will be clear when we look at the applications of SFRL for Channel Simulation and Lossy Source Coding Problems in the sections IV and V of the report.

On top of the applications of SFRL given in [1], an intuitive approach can be thought for applying SFRL to the privacy problems. Assume $Y$ to represent the random variable where the data that we want to send comes from, and assume $X$ to represent the random variable where the private part of this data comes from. Then, we can not directly send $Y$, because some attackers who knows $Y$, can reveal something that we want to keep private as $Y$ is not independent of $X$. Therefore, we want to send something comes from a random variable $Z$ which is independent of $X$, and we want to make this in a way that $Z$ is most informative about $Y$ to send the most of the information. Specifically, sending $Z$, the amount of information that we could not send about $Y$ is $H(Y|Z)$. Therefore, we can use SFRL to find a random variable $Z$ which we can use in our encoder to design the message that we will send without revealing any information about the private part of our data (since $Z$ is independent of $X$), and minimizing $H(Y|Z)$ that is the amount of information that we could not send about $Y$. The general framework is given in Fig. 1.

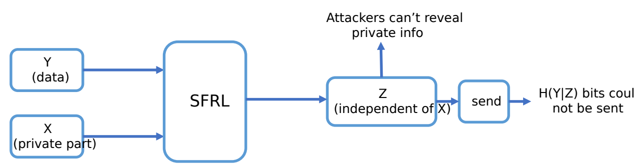


Figure 1. General framework for using SFRL for privacy applications.

Then, Poisson Matching Lemma and its applications to some one-shot achievability problems in Information Theory, which is introduced in [2], extends the work done in [1] by providing an upper bound for the probability of a random variable $\tilde{U}_Q$ given by the probability measure of $Q$ being different than a ***known*** random variable $\tilde{U}_P$ given by the probability measure of $P$ in terms of the density of these probability measures $P$ and $Q$ at the value of the random variable $\tilde{U}_P$ which is $Pr\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P\} \leq 1 - (1 + \frac{dP}{dQ}(\tilde{U}_P))^{-1}$. I am going to explain Poisson Matching Lemma with some examples and give the sketch of the proof in the section VI of this report.

Last but not least, I will talk about the goals of this project and what works I am planning to do to continue this project. I will mention the other applications of SFRL and Poisson Matching Lemma given in [1] and [2] and I will discuss other potential applications of them in the section VII.

## II. FUNCTIONAL REPRESENTATION

In this section, we will look at the functional representation problem in the general sense, and examine some natural limits.

*Definition 1 (Functional Representation Problem):* Let $X$ and $Y$ be given random variables. We want to find a random variable $Z$ which is independent of $X$ and a deterministic function $g$ such that $Y = g(X, Z)$ (i.e. $H(Y|X, Z) = 0$).

There are quite many approaches to these problem with different concerns. Let's examine the concern of minimizing the support size of $Z$ for the case where both $X$ and $Y$ are discrete random variables with finite supports (if one of them not, we need an infinite support for $Z$ anyways) first and natural limits on it. And then, let's continue with the concern of minimizing $H(Y|Z)$ and natural limits on it.

### A. Minimizing the support size of $Z$

Constructing a solution is quite easy in case where $X$ and $Y$ are discrete if we do not have any further constraints like minimizing the support size of $Z$. For example, we can choose $Z$ as a random vector of size $|\mathcal{X}|$ which is formed by independent random variables with the same distribution as $Y$. So, $Z$ will be independent of $X$ since it does not depend on which value that $X$ takes. Then, we can choose $g(X, Z)$ as the $i$th entry of the random vector $Z$ if $X = x_i$ where $x_i$ represents the $i$th entry of the support of $X$. This way, we could get $Y = g(X, Z)$. However, in this scheme, the support size of $Z$ which is $|\mathcal{Z}|$ equals to $|\mathcal{Y}|^{|\mathcal{X}|}$ because $Z$ is a random vector of size $|\mathcal{X}|$ and each element of this random vector has a support size of $|\mathcal{Y}|$.

This brings us the question of "what is the minimum support size for $Z$?". As stated in [1], the smallest possible support size for $Z$ equals to $|\mathcal{X}|(|\mathcal{Y}| - 1) + 1$. We can construct a $|\mathcal{Z}|$ and the corresponding function $g$ that achieves this support size as follows. Let

$$A_{ij} = \sum_{k=1}^{j} Pr[Y = y_k|X = x_i]$$

for all $i = 1, 2, ..., |\mathcal{X}|$ and $j = 1, 2, ..., |\mathcal{Y}| - 1$. Next, let $q_k$ be the $k$th smallest element of $A_{ij}$ for $k = 1, 2, ..., |\mathcal{X}|(|\mathcal{Y}| - 1)$ and let $q_{|\mathcal{X}|(|\mathcal{Y}|-1)+1} = 1$. Additionally, let's define $p_1 = q_1$ and $p_i = q_i - q_{i-1}$ for $i = 2, 3, ..., |\mathcal{X}|(|\mathcal{Y}| - 1) + 1$. Notice that $\sum_{i=1}^{|\mathcal{X}|(|\mathcal{Y}|-1)+1} p_i = 1$. Then, we choose $Z$ as a random variable that takes value $z_i$ with probability $p_i$. As we can see $Z$ does not depend on $X$, but it only depends on the conditional distribution $P_{Y|X=x_i}$ for $i = 1, 2, ..., |\mathcal{X}|$. Moreover, we can design the function $g$ such that $g(x_i, z_j) = y_k$ such that

$$k = \min_{l=1,2,...,|\mathcal{Y}|} \{l : \sum_{m=1}^{l} Pr[Y = y_m|X = x_i] \geq q_j\})$$

Then, $g(X, Z) = Y$ because

$$Pr[X = x_i, Y = y_j] = Pr[X = x_i, g(X, Z) = y_j]$$

for all $i = 1, 2, ..., |\mathcal{X}|$ and $j = 1, 2, ..., |\mathcal{Y}|$.

Therefore, this scheme solves the functional representation problem with using support size at most $|\mathcal{Z}| = \mathcal{X}|(|\mathcal{Y}|-1)+1$ which is the minimum required support size for $Z$ as stated in [1].

### B. Minimizing $H(Y|Z)$

Let's start with an example.

*Example 1:* Let $B_1$, $B_2$, and $B_3$ be i.i.d random variables with equiprobable Bernoulli Distribution, and let $X$ be $(B_1, B_2)$ and let $Y$ be $(B_2, B_3)$. Then, both $Z_1 = B_3$ and $Z_2 = B_1 \oplus B_3$ can be used for functional representation. However, using $Z_2$ is actually redundant because $Z_1$ is more informative about $Y$. To see it more clearly, let's calculate $H(Y|Z_1)$ and $H(Y|Z_2)$.

$$H(Y|Z_1) = H(B_2, B_3|B_3) = H(B_2) = 1$$

$$H(Y|Z_2) = H(B_2, B_3|B_1 \oplus B_3) = H(B_2, B_1) = 2$$

Therefore, we can see that $Z_1$ is more informative about Y.

Now, it is natural to ask that if there exists a lower bound for $H(Y|Z)$. To explore:

$$H(Y|Z) = \underbrace{H(Y|X,Z)}_{=0 \text{ by } g(X,Z)=Y} + I(X,Y) + \underbrace{I(X,Z|Y)}_{\geq 0} - \underbrace{I(X,Z)}_{=0 \text{ by } X \perp Z}$$

Therefore, $I(X, Y)$ is a natural limit that we can not surpass for minimizing $H(Y|Z)$. Now, let's introduce our almost perfect tool SFRL [1] that provides functional representation with an upper bound for $H(Y|Z)$ which is close to the natural lower bound of $I(X, Y)$. Let's introduce it in the next chapter.

## III. Strong Functional Representation Lemma

Here is the statement of SFRL which is the main theorem of [1].

*Theorem 1 (Strong Functional Representation Lemma):* Given random variables $X$ and $Y$, we can find a random variable $Z$ and a function $g$ such that:

$$-X \perp Z$$

$$-Y = g(X, Z)$$

$$-H(Y|Z) \leq I(X, Y) + \log(I(X, Y) + 1) + 4$$

Thus, SFRL provides a very good bound bound of order $O(n + log(n))$ for the natural limit $n = I(X, Y) \leq H(Y|Z)$.

Now, let's examine the construction for $Z$ and $g$ suggested in [1]. $Z$ is given as the Poisson point process as follows:

$$Z = \{(T_i, \tilde{Y}_i)\}_{i=1}^{\inf}$$

where $T_i$'s are the event times of a Poisson process with rate 1 and $\tilde{Y}_i$'s are i.i.d random variables distributed as $Y$. Then, $Z$ is clearly independent of $X$ by its definition.

Moreover, we can choose $g(X, Z)$ as

$$g(x, z) = g(x, \{\tilde{y}_i, t_i\}) = \tilde{y}_{k(x,\{\tilde{y}_i,t_i\})}$$

where

$$k(x, \{\tilde{y}_i, t_i\}) = \arg\min_i t_i \frac{dP_Y}{dP_{Y|X=x}}(\tilde{y}_i)$$

Now, let's explain why this scheme works as a functional representation for the discrete case.

*Proof:* (for functional representation part only)

Assume $\mathcal{Y} = \{y_1, y_2, ..., y_{|\mathcal{Y}|}\}$. Then, let's imagine $|\mathcal{Y}|$ many Poisson point processes with rates $P_Y(y_i)$'s and labels

$y_i$'s for $i = 1, 2, ..., |\mathcal{Y}|$. Combination of all of these Poisson Processes will make another Poisson process with rate $\sum_{i=1}^{|\mathcal{Y}|} P_Y(y_i) = 1$ due to the properties of Poisson processes. So, let's consider $T_i$'s as the event times of this Poisson process with rate 1.

Now, let's also consider that we strecth all of these Poisson processes with $\frac{dP_Y}{dP_{Y|X=x}}(\tilde{y}_i)$ respectively for all $i = 1, 2, ..., |\mathcal{Y}|$. Then, we will acquire again $|\mathcal{Y}|$ many Poisson processes, but this time with rates $P_{Y|X=x}(y_i)$'s and labels $y_i$'s for $i = 1, 2, ..., |\mathcal{Y}|$.

Then, finding the $k(x, \{\tilde{y}_i, t_i\}) = \arg\min_i t_i \frac{dP_Y}{dP_{Y|X=x}}(\tilde{y}_i)$ means finding the label of the first event time of the Poisson process acquired as the combination of all streched versions of the Poisson processes that we started with. Due to the Poisson process properties, it can be found as $\frac{P_{Y|X=x}(y_i)}{\sum_{i=1}^{|\mathcal{Y}|} P_{Y|X=x}(y_i)} = \frac{P_{Y|X=x}(y_i)}{1} = P_{Y|X=x}(y_i)$. Therefore, we have:

$$Pr[\tilde{Y}_k = y_i | X = x] = P_{Y|X=x}(y_i)$$

for all $x \in \mathcal{X}$. Therefore:

$$Pr[\tilde{Y}_k = y_i | X] = P_{Y|X}(y_i)$$

So, $g(X, Z) = \tilde{Y}_k = Y$. ■

Moreover, this construction also satisfies:

$$H(Y|Z) \leq I(X;Y) + log(I(X;Y) + 1) + 4$$

The proof of this information theoretic upper bound is given in [1], but I will skip it in this report because it is quite lengthy and complicated. However, it is good to note that, by examining and altering some of the parts of the proof given in [1], we saw that $H(Y|Z)$ can be optimized by 3 more bits (i.e. $H(Y|Z) \leq I(X;Y) + log(I(X;Y) + 1) + 1$).

Now, let's analyze this upper bound given for the conditional entropy $H(Y|Z)$. First, we can say that this is a very good upper bound for $H(Y|Z)$ considering that $I(X;Y)$ is a lower bound that can not be overcome. Second, we can say that, this is an almost (up to 5 bits) perfect bound because it is shown in [1] that for a fixed $I(X;Y)$, we can not guarantee to come up with a $Z$ satisfying functional representation and $Z \perp X$ condition with $H(Y|Z) \leq I(X;Y) + log(I(X;Y) + 1) - 1$.

Now, let's move to the applications of SFRL.

## IV. One-Shot Channel Simulation

One of the most useful application of SFRL is for the One-Shot Channel Simulation problem. The framework for the problem is provided in Fig. 2 below.

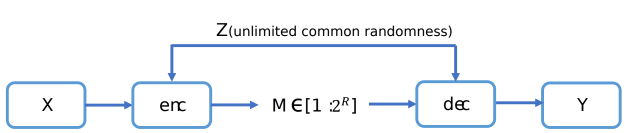


Figure 2. One-Shot Channel Simulation Scheme

We basically want to create a conditional distribution $P_{Y|X}$ by designing an encoder and decoder functions that are allowed to share unlimited common randomness Here is the exact statement for the problem:

*Definition 2 (One-Shot Channel Simulation Problem):* For given random variables $X$ and $Y$, we want to simulate a channel $P_{Y|X}$ by designing an encoder, decoder pair ([enc, dec]) with unlimited common randomness $Z$ such that we can represent the message $M$ (which is the output of the encoder and the input of the decoder) in as small number of bits as possible. If we can do that in $R$ bits, then $R$ is called as an achievable rate.

It turns out that an achievable rate $R$ should be at least the capacity of the channel $C(P_{Y|X}) = \max_{P_X} I(X;Y)$ by the theorem given by Bennett, Shor, Smolin, and Thapliyal in [3] stated below:

*Theorem 2 (BSST Theorem, 2002):* Suppose that $\inf\{R :$ $R$ is an achievable rate for the channel simulation problem$\}$ is called as $R^*_{\text{channel simulation}}$. Then:

$$R^*_{\text{channel simulation}} = \max_{P_X} I(X;Y)$$

Now, let's see how we can approach this problem by utilizing SFRL.

We want to simulate a channel $P_{Y|X}$ for a given input $X$. Since $X$ and $P_{Y|X}$ is known, $Y$ is known as well. Then, for $X$, and $Y$, we can acquire a $Z$ s.t. $Z \perp X$, $H(Y|X,Z) = 0$, and $H(Y|Z) \leq I(X;Y) + \log(I(X;Y) + 1) + 4$. The idea is using this $Z$ as the common randomness between encoder and decoder.

Now, since we are using the $Z$ acquired by SFRL as the common randomness, we can argue the following. First, since encoder knows $X$ and $Z$, it also knows $Y$ as $Y$ can be represented as a deterministic function of $X$ and $Z$. Then, it knows $P_{Y|Z}$. Now, using the message $M$, the encoder can tell $P_{Y|Z}$ to the decoder by $\lceil H(Y|Z) \rceil$ bits using Huffmann Coding. Then, decoder receives $P_{Y|Z}$ and it can recover $Y$ by using it together with the common randomness $Z$. Then, $\lceil H(Y|Z) \rceil$ is an achievable rate. Since this common randpomness $Z$ is acquired by SFRL, we know:

$$\lceil H(Y|Z) \rceil < H(Y|Z) + 1 \leq I(X;Y) + \log(I(X;Y) + 1) + 5$$

$$\leq R^*_{\text{channel simulation}} + \log(R^*_{\text{channel simulation}} + 1) + 5$$

Thus, using SFRL, we can find an achievable rate which is almost as good as the natural limit $R^*_{\text{channel simulation}}$. This one shot result given in [1] gives better results than the one-way protocol given in [4] which provides an upper bound of for the asymtotical case. So, SFRL is quite useful for the channel simulation problem compared to the other techniques in the literature as it provides one-shot results even better than the results of the other techniques for the asymtotical case.

## V. Lossy Source Coding

Another useful application of SFRL is for the Lossy Source Coding problem. Here is the exact statement for the problem:

*Definition 3 (Lossy Source Coding Problem):* For a given random variable $X$, we want to design an encoder, decoder pair ([enc, dec]) such that $E[d(X,Y)] \leq D$ where $Y$ is the random variable that represents the output of the decoder. We want to design this ([enc, dec]) such that we represent the message $M$ (which is the output of the encoder and the input of the decoder) in as small number of bits as possible. If we can do that in $R$ bits, then $R$ is called as an achievable rate. Moreover, for a given random variable $X$ and a fixed distortion function $d$, we define $R(D) = inf\{R : R \text{ is achievable}\}$. The framework is given in Fig. 3.

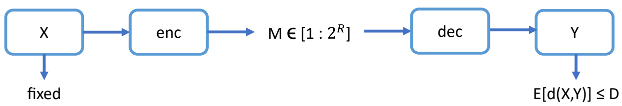


Figure 3. Lossy Source Coding Scheme

The natural lower bound for the achievable rate of Lossy Source Coding problem is given by the Lossy Source Coding Theorem as follows:

$$R(D) = \min_{P_{Y|X}:E[d(X,Y) \leq D]} I(X;Y)$$

Again by the Lossy Source Coding Theorem, this limit is achievable. However, the typicality ([enc, dec]) is used in the proof and it is not implementable as it requires to construct and store an $n \times 2^{nR(D)}$ dimensional random matrix as the codebook.

SFRL can be used to design an ([enc, dec]) pair which is easy to implement and achieves a rate of $\hat{R}$ for all $\hat{R} \geq R(D) + \log_2(R(D)+1)+6$ as shown in [1]. Therefore, we can actually use SFRL to design a Lossy Source Coding Scheme with a reasonable rate (in the order of $O(R(D)+\log_2(R(D)))$ for the natural lower limit $R(D)$). Moreover, the design given in [1] is almost deterministic, so it does not use a random encoder unlike the typicality encoder that achieves $R(D)$. The reason we call this design as almost deterministic is that it actually uses only 1-bit of randomness for codebook generation in the proof.

## VI. Poisson Matching Lemma

Now, let's present the Poisson Matching Lemma given in [2]. It is built on top of the work in [1] and its idea is based on the properties of Poisson Point Processes just like SFRL.

Now, let's introduce the Poisson Matching Lemma which is the main theorem of [2]:

*Theorem 3 (Poisson Matching Lemma):* Let $\{\hat{U}_i, T_i\}_{i\in N}$ be the points of a Poisson process with intensity measure $\mu$ (Lebesque Measure). Also, let $P$ and $Q$ be some probability measures on $\mathcal{U}$ with $P, Q << \mu$ (i.e. absolute continuity). Then, $\tilde{U}_P$ and $\tilde{U}_Q$ are defined as:

$$\tilde{U}_P(\{\hat{U}_i, T_i\}_{i\in N}) := \hat{U}_{K_P(\{\hat{U}_i,T_i\}_{i\in N})}$$

$$\tilde{U}_Q(\{\hat{U}_i, T_i\}_{i\in N}) := \hat{U}_{K_Q(\{\hat{U}_i,T_i\}_{i\in N})}$$

where

$$K_P(\{\hat{U}_i, T_i\}_{i\in N}) := \underset{i:\frac{dP}{d\mu}(\hat{U}_i)>0}{\text{argmin}} \; T_i \frac{dP}{d\mu}(\hat{U}_i)^{-1}$$

$$K_Q(\{\hat{U}_i, T_i\}_{i\in N}) := \underset{i:\frac{dQ}{d\mu}(\hat{U}_i)>0}{\text{argmin}} \; T_i \frac{dQ}{d\mu}(\hat{U}_i)^{-1}$$

with an arbitrary tie-breaking that happens with zero probability. Moreover, for the sake of easier notation, we will notate $\tilde{U}_P(\{\hat{U}_i, T_i\}_{i\in N})$ as simply $\tilde{U}_P$ if $\{\hat{U}_i, T_i\}_{i\in N}$ is clear from the context.

Then, we have the following inequality almost surely:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P\}) \leq 1 - \frac{1}{1 + \frac{dP}{dQ}(\tilde{U}_P)}$$

*Proof:* (for discrete alphabets $\mathcal{U}$)

Let $Z_u \sim \text{Exp}(1)$ be i.i.d random variables. Then, we can represent $\tilde{U}_P$ as $\text{argmin}_u Z_u/P(u)$ by the properties of Poisson processes when we think $Z_u$'s as the event times for a Poisson process with rate 1. Let $\tilde{Z}_P := \frac{Z_{\tilde{U}_P}}{P(\tilde{U}_P)}$.

Conditional on $\tilde{U}_P = u$ and $\tilde{Z}_P = \tilde{z}$, we know $\frac{Z_v}{P(v)} \tilde{z}$ for all $v \in \mathcal{U}$ different than $u$. Then, by the memoryless property of exponential distribution, we have $Z_v - P(v)\tilde{z} \sim \text{Exp}(1)$ because $P(Z_v > c) = P(Z_v - P(v)\tilde{z} > c | Z_v P(v)\tilde{z})$ for any arbitrary c due to memoryless property and $Z_v$ is known to be exponentially distributed with rate 1. Moreover, $Z_v - P(v)\tilde{z}$'s are independent across $v$'s and independent of $\tilde{Z}_P$. The first independency claim follows due to the independency of $Z_v$'s across $v$'s. Here, v is an any element in $\mathcal{U}$ different than $u$, therefore for $u \in \mathcal{U}$, we see $Z_v - P(v)\tilde{z}$'s are independent of $\tilde{Z}_P$ due to the independency of $Z_v$'s with $Z_u$.

Now, using all these observations, let's prove the inequality given in the statement of the Poisson Matching Lemma:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\}) = \Pr(\{\min_{v\neq u} \frac{Z_v}{Q(v)} \neq \frac{Z_u}{Q(u)} | \tilde{U}_P = u\})$$

$$\leq \Pr(\{\min_{v\neq u} \frac{Z_v - P(v)\tilde{Z}_P}{Q(v)} \neq \frac{\tilde{Z}_P P(u)}{Q(u)} | \tilde{U}_P = u\})$$

$$= \Pr(\{\text{Exp}(1 - Q(u)) \leq \text{Exp}(\frac{Q(U)}{P(u)}) | \tilde{U}_P = u\})$$

for independent exponential distributions. The last line is because $Z_v - P(v)\tilde{Z}_P$'s are independent exponential distributions with rate 1, thus $\frac{Z_v - P(v)\tilde{Z}_P}{Q(v)}$'s are independent exponential distributions with rate $Q(v)$'s, then $\min_{v\neq u} \frac{Z_v - P(v)\tilde{Z}_P}{Q(v)}$ is an exponential distribution with rate $\sum_{v\neq u} Q(v) = 1 - Q(u)$. This is because finding the minimum over $\frac{Z_v - P(v)\tilde{Z}_P}{Q(v)}$'s is like finding the first event time of the combination of some Poisson processes with rates $Q(v)$'s. Thus, the combination becomes another Poisson process with rate $\sum_{v\neq u} Q(v)$ and the first event time of this Poisson process (which is

$\min_{v\neq u} \frac{Z_v - P(v)\tilde{Z}_P}{Q(v)}$) is an exponential distribution with rate $\sum_{v\neq u} Q(v)$.

Then, we have got the following inequality:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\})$$
$$\leq \Pr(\{\text{Exp}(1 - Q(u)) \leq \text{Exp}(\frac{Q(U)}{P(u)}) | \tilde{U}_P = u\})$$

Here, we can get rid of the condition for the right hand side of the inequality because we already reduced the problem to comparing independent exponential distributions. Then:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\}) \leq \Pr(\{\text{Exp}(1-Q(u)) \leq \text{Exp}(\frac{Q(U)}{P(u)})\})$$

$$= \frac{1 - Q(u)}{1 - Q(u) + \frac{Q(u)}{P(u)}} \leq \frac{1}{1 + \frac{Q(u)}{P(u)}} = 1 - \frac{1}{1 + \frac{P(u)}{Q(u)}}$$

■

Note that Poisson Matching Lemma can be applied to a lot of problems in Information Theory such as various setting of Channel Simulation and Lossy Source Coding Problems. These applications are listed as channels with state information at the encoder, lossy source coding with side information at the decoder, joint source-channel coding, broadcast channels, distributed lossy source coding, multiple access channels and channel resolvability in [2]. We will not examine these applications one by one in these report, but instead we will look at the some other potential applications of SFRL and Poisson Matching Lemma to new problems in the next section.

## VII. Future Directions

One of my main goal in this project is to understand the applications of SFRL and Poisson Matching Lemma given in [1] and [2]. Up to this section, we already discussed some applications of these theorem including One-Shot Channel Simulation in IV and Rate Distortion in V. To take this project further, I will continue with the other important applications of these theorems and I will consider about the application of these theorems to potential new problems.

### *A. Other Application Given in [1] and [2]*

1) One-Shot Channel Coding ([2], section III)
   In [2], Poisson Matching Lemma is applied to the one-shot channel coding problem given in [5] to have a bound. The encoder takes an integer $M$ which is an uniformly distributed random variable between integers 1 to $L$, and maps to $X$. Then, $X$ is sent through the channel $P_{Y|X}$ and $Y$ is the output of the channel. The decoder observes Y and recovers $\hat{M}$ with error probability $P_e = \Pr\{M \neq \hat{M}\}$. Then, the following lemma is stated in [2]:
   *Lemma 4:* Fix a $P_X$. If $P_{XY} << P_X \times P_Y$, then there exists a code for the channel $P_{Y|X}$, with message $M \sim$ Unif$[1 : L]$, and with average error probability
   $$P_e \leq \text{E}[1 - (1 + L2^{\imath(X;Y)})^{-1}]$$
   where $\imath(X;Y)$ represents the information density defined as:
   $$\imath(X;Y) := \log \frac{d(P_{XY})}{d(P_X \times P_Y)}(x,y)$$
2) One-Shot Channel Resolvability and Soft Covering ([2], section XII)
   The one-shot channel resolvability setup is given in [6] as follows:
   Consider $P_{Y|X}$ and $P_X$ as given. The encoder takes an integer $M$ which is an uniformly distributed random variable between integers 1 to $L$, and maps to $g(M) = \hat{X}$. Then, $\hat{X}$ is sent through the channel $P_{Y|X}$ and $\hat{Y}$ is the output of the channel. The aim is to minimize the total variation between $P_{\hat{Y}}$ and $P_Y$.
   In proposition 3 of ([2], section XII), it is proven that if the expected total variation between $P_{\hat{Y}}$ and $P_Y$ is less than for some given upper bounds, then there exists a code for channel resolvability.

### *B. Potential New Problems*

1) Universal Lossless Source Coding
   It is the compression problem where we do not know the distribution of the input beforehand. We have already seen the applications of SFRL for the one-shot channel simulation and lossy source coding problems where we used Huffman coding as we have the assumption of that the input distribution is known. This time, we can use Lempel-Ziv or other universal coding techniques to apply SFRL to Universal Source Coding Problem.
2) Secret Key Generation from Joint Randomness
   Secret Key Generation Problem from Common Randommness is introduced in [7]. Note that this work is way before than [1] and [2]. We are thinking that SFRL might be a useful tool for this problem since it allows us to create a random variable $Z$ that does not contain any information about $X$, therefore it can be used as a Secret Key, and it is still quite informative about $Y$. I will investigate ways to use SFRL for the Secret Key Generation Problem and compare them with the methods used in [7].

In conclusion, I studied SFRL and Poisson Matching Lemma given in [1] and [2] with some of their application in this project so far. I am planning continue studying with other applications of these theorems to some important theorems in Information Theory discussed in VII-A and I am planning to study new problems discussed in VII-B and, hopefully, get some meaningful results for these problems utilizing SFRL and Poisson Matching Lemma.

## References

[1] C. T. Li and A. E. Gamal, "Strong functional representation lemma and applications to coding theorems," *IEEE Transactions on Information Theory*, vol. 64, no. 11, p. 6967–6978, 2018.
[2] C. T. Li and V. Anantharam, "A unified framework for one-shot achievability via the poisson matching lemma," *IEEE Transactions on Information Theory*, vol. 67, no. 5, p. 2624–2651, 2021.

[3] C. Bennett, P. Shor, J. Smolin, and A. Thapliyal, "Entanglement-assisted capacity of a quantum channel and the reverse shannon theorem," *IEEE Transactions on Information Theory*, vol. 48, no. 10, pp. 2637–2655, 2002.

[4] P. Harsha, R. Jain, D. McAllester, and J. Radhakrishnan, "The communication complexity of correlation," *IEEE Transactions on Information Theory*, vol. 56, no. 1, pp. 438–449, 2010.

[5] M. H. Yassaee, M. R. Aref, and A. Gohari, "A technique for deriving one-shot achievability results in network information theory," *2013 IEEE International Symposium on Information Theory*, 2013.

[6] T. Han and S. Verdu, "Approximation theory of output statistics," *IEEE Transactions on Information Theory*, vol. 39, no. 3, pp. 752–772, 1993.

[7] R. Ahlswede and I. Csiszar, "Common randomness in information theory and cryptography. i. secret sharing," *IEEE Transactions on Information Theory*, vol. 39, no. 4, p. 1121–1132, Jul 1993.